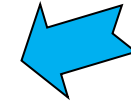


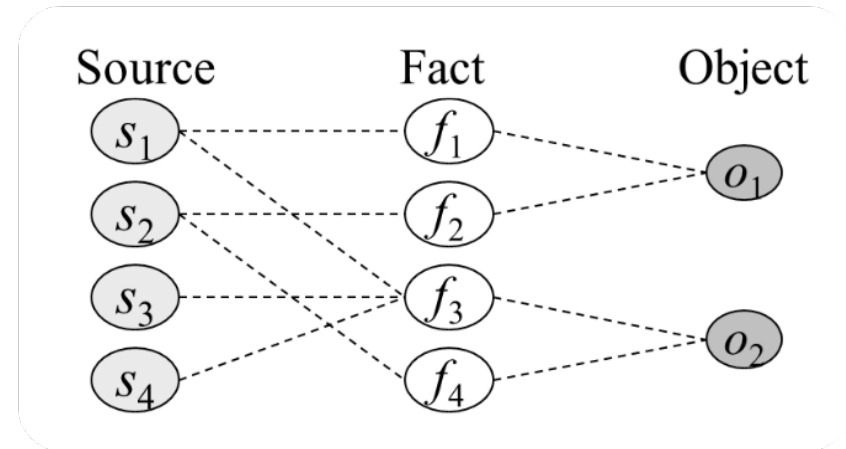
Data Mining Applications

- ❑ Data Mining for Sentiment and Opinion
- ❑ Truth Discovery and Misinformation Identification
- ❑ Information and Disease Propagation
- ❑ Productivity and Team Science



Truth Discovery

- ❑ Goal: assess and choose the actual true value for a data item when different sources provide conflicting information on it.
- ❑ Single-truth discovery:
 - ❑ Only one true value is allowed for data item.
- ❑ Multi-truth discovery:
 - ❑ Multiple true values are allowed for data items.
- ❑ Challenges:
 - ❑ Source trustworthiness and claim reliability.
 - ❑ Limited amount of labeled data source: semi-supervised/unsupervised.
 - ❑ Detect copy behaviors of false values.
 - ❑ Evaluation: based on data distribution and treat those deviated substantially from norm as wrong answers.



Identification of Misinformation

- ❑ Goal: detect false, inaccurate, or misleading information from data source.
- ❑ Potential solutions:
 - ❑ Explore mutual enhancement between source trustworthiness and claim credibility (similar idea from truth discovery).
 - ❑ Methods should distinguish misinformation with additional measures:
 - ❑ Building up information literacy and media literacy through education.
 - ❑ Using commonsense knowledge and open-mindedness.
 - ❑ Developing critical thinking.
 - ❑ Determine misinformation in the final stage based on:
 - ❑ Individual's mental model and worldview beliefs.
 - ❑ Consideration of repetition of misinformation.
 - ❑ Time-lag/relative coherency between misinformation and corrective information.